Q. What is the difference between Routing and Route protocols?

→ * Routing Protocols

- Definition

A routing protocol is a protocol used by routers to exchange routing information and build routing tables.

- Purpose

It helps routers
- Discover remote networks
- Calculat best path
- Update routing table manually
- React to topology changes.

* Control Plane Vs Data Plane

Routing protocol = Control plane
Route Protocol = Data plane
control plane Decides
Data plane delivers.

- Work Between Routers ↔ Routers

* Examples ~~eg~~.

- RIP -- Routing Information Protocol
- OSPF - Open Shortest Path First
- EIGRP - Enhanced Interior Gateway Routing protocol
- BGP - Border Gateway Protocol.

* Note.

Routing protocols do not carry user data, they only exchange routing information.

* Route Protocol

- Definition

A routed protocol is protocol that carries actual user data packets across network.

- Purpose

It provides:
- logical addressing
- Packet structure
- End to end delivery

→ works between:

End device ↔ Routers ↔ End devices

- Exaples
- Internet Protocol (IP)
- Internetwork packet exchange (IPX)

* Key point

Routed protocols are forwaded by routers.

* Conceptual Flow

Routing Protocol - Builds routing table
Routing Table - Used to forword
Routed Protocol - Actual packet being Forwarded

* Comparison

Feature	Routing Protocol	Route Protocol
Purpose	Exchange route information Builds Routing table	Carries user data
Used by	Routers Only	End devices + Routers
Data type	Control plane traffic	Data plane traffic
Builds Routing Table?	Yes	No
Forwarded by Router?	Usually not	Yes
Example	OSPF, RIP, BGP	IP

* Edge Cases

1) Static Routing Scenario

If only static routes are config:
- No routing protocol is used
- Routed protocol (IP) still operates

Edge Insight
Routing can exist without a routing protocol.

2) Single Network Environment

IP only one subnet
- No routing required
- Routed protocol still operates.

3) Routing Protocol Carries its Own Packets

Ex:
- OSPF packets use IP as transport
- BGP uses TCP (port 179)

So routing protocol themselve use a route protocol to f^n?

~~* Vulnerabilities~~
Routing ~~protocols~~ are a ~~major~~ attack ~~surface~~

4) Default Route Only

Router with only

ip route 0.0.0.0 0.0.0.0 <nxt-hope>

No dynamic routing protocol
still forwarding routed protocol traffic.

* Vulnerabilities

Routing protocols are major attack surface

1) Routing Protocol Spoofing

Attacker injects fake routes

Ex:

- Fake OSPF LSA
- Rogue RIP updates
- BGP route hijacking

Impact

- Traffic redirection
- Man-in-the-middle
- Blackhole attack

BGP hijacks are global level incidents

2) Route Table Overflow

Attacker floods routing updates

Effect

- CPU exhaustion
- memory exhaustion
- Router crash

3) BGP Hijacking

Malicious AS advertieses prefixes it doesn't own.

Effect: Internet scale traffic interception.

4) Route Protocol Exploits

Example

- IP Fragmentation attack
- TTL manipulation
- ICMP abuse
- ARP poisoining

Impact

- DoS
- Session hijacking

* Mitigation Strategies

a) For Routing Protocol Security

1) Authentication

OSPF: ip ospf authentication message-digest

BGP:
- MD5 authentication
- TTL security check

2) Route Filtering

Usse:

- Prefix-lists
- Route-maps
- Distribute-lists

Prevent rogue advertisement

3) Max Prefix limit (BGP)

Prevent memory exhaustion: neighbor x.x.x.x maximum-pefix

4) Control Plane Policies (CoPP): Limit control-plane traffic rate

5) Use Passive Intefaces: Prevents unwanted neighbor formating.

b) For Routed Protocol Security

1) ACL Implementation: Block authorized traffic

2) Anti-spoofing filters: Ingress filtering (BCP38 principle)

3) Disable ~~Usuable~~ Unused protocol: Disable IP redirects, Proxy ARP if unnecessary

A routing protocol operates in the control plane to exchange and compute network path information, while a routed protocol operates in the data plane to encapsulate and transport end-user traffic across those computed paths.

* Can a router forward packets without a routing protocol?
Yes - using static routes or directly connected routes.

Q.2 Boot Process

Booting is the controlled sequence that transitions a system from powered-off hardware state to fully operational OS env+

stages:

1) Hardware intialization
2) Firmware execution
3) Boot loader stage
4) Kernel loading
5) User-space initialization
6) ~~Edge cases~~

stage
(1) Power on & Hardware initialization

— Power supply stabilization
- PSU performs voltage regulation
- CPU reset vector is triggered
- Execution begins at firmware-defined memory location

— POST (Power-on Self Test)

Performed by firmeware (BIOS/UEFI):
- RAM check
- CPU verification
- GPU detection
- Peripheral enumeration
- keyboard controller test
- Storage Device detection

If failure Occurs → beep codes or error screen

tage
(2) Firmware (BIOS vs UEFI)

legacy BIOS
- 16 bit real mode.
- Reads first 512 bytes from boot disk (MBR)
- Limited hardware abstraction.

Modern UEFI
- 32/64-bit mode
- Has built-in boot manager
- Uses EFI system partition (ESP)
- Support Secure Boot

Stage

③ Bootloader

Bootloader loads the OS kernel into memory

In linux • GRUB

In window • Windows Boot manager

Boot loader responsibilities
- Select OS (dual boot scenario)
- load kernel image
- load initramfs / initrd
- Pass kernel parameters

Stage

④ Kernel intialization

Onece kernel loads
1) Switches CPU to protected / long mode
2) Initializes memory management
3) Mounts root filesystem
4) Initializes drivers
5) Starts first process

In linux • PID1 → systemd

Kernel set up:
- Process scheduler
- Interrupt handling
- Networking stack
- File system

Stage

⑤ User space ~~ui~~ initialization

system services start
- Networking daemon
- Display manager
- Login service
- SSH server

system reaches
- Multi - user mode
- Graphical mode (if desktop)

Boot Flow

Power on
↳ POST
↳ firmware
↳ Bootloader
↳ kernel
↳ Init system
↳ services
↳ login prompt

Edge cases

1) Carrupted Bootloader
- System stuck at firmware
- "No bootable device found

Fix
- Boot from recovery USB
- Reinstall GRUB/ Repair window boot

2) MBR vs GPT mismatch
legacy BIOS cannot boot GPT without compatibility mode

3) Secure Boot Failure
Unsigned kernel or driver blocked by UEFI secure Boot

4) Kernel Panic
occurs If
- Root filesystem not found
- Incompatible driver
- memory corruption
-

Linux shows: kernel panic - not syncing

5) Boot loop
Causes:
- Bad driver
- Faulty update
- Malware
- Corrupted system file

* Security Vulnerabilities in Boot process

Boot phase is extremely sensitive because OS-level security is not fully active yet.

1) Boot kit | Rootkit Attack

malware infects
- MBR
- EFI partition
- Bootloader

Ex
- Mebroot
- LoJax

These load before OS → extreamely stealthy

2) Evil Maid Attack

Attacker with physical access.
- Modifices bootloader
- Installs keylogger
- Extracts disk encryption key
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Mitigation
- Full disk encryption
- Secure Boot
- TPM

3) Secure Boot Bypass

If secure boot disabled
- Unsigned OS can load
- Malware persists

uses:
- Trusted platform Module (TPM)

4) Firmware Exploits

UEFI vulnerabilities allows
- Persistent malware
- Survive OS reinstall

Firmeware rarely updated puts high value target

* Defensive Controls

1) Enable Secure Boot
2) Enable TPM
3) Use Full Disk Encryption
4) Disable Extrenal boot devices
5) Update firmware regularly
6) Set BIOS/UEFI Password

Boot process is multi-statge initialization sequence involvi firmware execution, bootloader loading, kernel intialization and service startup. It is high-risk attack surface due to pre-os execution privileges, making secure boot mechanisms and firmware integrity critical.

Q.3 Memory in Router

Routers are specialized embedded systems. their memory architecture is optimized for deterministic packet forwarding. fast lookup operations and configuration persistence.

- Router memory architecture consists of volatile (RAM) and non-volatile (ROM, NVRAM, flash) components that support os execution, configuration persistence and packet forwarding.
- Memory exhaustion, corruption or exploitation can result in denial of service, configuration loss or sys compromise
- Mitigation requires resource control, firmware updates and strict access management.

Memory	Volatile	Stores
RAM	Yes	IOS (running), active config running Routing table, buffers
NVRAM	No	Startup config, conf register
Flash	No	IOS image, Backup IOS
ROM	No	POST, bootstrap ROMMON (ROM Moniter Mode

Q.4) Modes of Router

Router "modes" refer to operational privilege contexts in the CLI. each mode controls
- what commands are permitted.
- what system components can be modified
- what level of authority the user has

Mode	Prompt	Risk level	Purpose
User EXEC	>	Low	* monitoring - lowest privilege level - Basic monitoring commands only - no configuration changes allowed (ping, sh ip interface brief, traceroute, sh ver)
Privileged EXEC	#	High	* full control - full monitoring access - can enter configuration mode - can save configuration - reload device (reload, debug, copy run-config, erase startup config)
Global config	(config)#	High	* Device config - modify device-wide settings - root of configuration hierarchy (hostname, ip routing, router ospf, line console)
Interface	(config-if)#	medium	* interface config - used to configure ↳ IP add. ↳ shdown / notshutdown ↳ speed / duplex ↳ ACL application
Line	(config-line)#	High	* Remote access used for ↳ console ↳ VTY (Telnet/SSH) ↳ AUX port
Router	(config-router)#	High	* Routing configuration / Routing protocol mode used for ↳ OSPF ↳ EIGRP ↳ BGP
ROMMON	rommon>	Critical	* Recovery - Password recovery - IOS recovery - manual boot